

Digital Edify Agentic CRM

Customer Data Foundation — Architecture at a Glance

Audience: Founders, business leadership, product, sales, and partners evaluating how our CRM stores and reasons about people. **Status:** Live. This is what the platform runs on today, not a proposal. **One-line summary:** We put every person and organization the business ever touches into one unified record — modelled on the same principles ServiceNow and Salesforce use — and the entire product operates from that single source of truth.

1. Why this matters — in one page

Every CRM eventually answers the same question: *who is this person, and what is our full history with them?*

Most CRMs answer it partially. A lead is one record, the same person after they enrol becomes a different record, an employee who later takes a course is a third record, and the same prospect who reaches out from two channels becomes duplicates. That's why advisors call the same lead twice, why WhatsApp campaigns go to people who never opted in, and why "show me everything about this person" is a five-tab exercise.

We fixed this by building the platform on a **single customer data foundation** — one record per real-world actor — with:

- **One identity** for every person or organization
- **Multiple ways to reach them** (emails, phones, addresses, WhatsApp) tracked per channel
- **State that changes over time** (lead → customer → alumnus) without breaking history
- **Consent recorded per channel** to satisfy DPDP (India) and GDPR
- **Automatic duplicate detection** with human-review before merging
- **Full audit trail** on every merge and every consent change

This is the same foundation ServiceNow uses to run enterprise support at Fortune 500 scale, and Salesforce uses to run the world's largest CRM. We took the best of both.

```

    flowchart TB
      subgraph identity["How we recognise them"]
        CP[Contact methods<br/>emails, phones, addresses<br/>each verified & consented]
        EX[External IDs<br/>Instagram, payment gateway,<br/>login provider]
      end

      subgraph state["Their journey"]
        PR[Roles over time<br/>lead → learner → alumnus<br/>full history preserved]
        AFF[Organization membership<br/>who works where,<br/>in what capacity]
        PAR[Company hierarchy<br/>subsidiaries, groups]
      end

      subgraph consent["What we can send them"]
        PC[Consent per channel<br/>WhatsApp · email · SMS · calls<br/>opt-in / opt-out / ev]
      end

      subgraph merge["Keeping data clean"]
        RULE[Match rules<br/>4 configurable strategies]
        CAND[Review queue<br/>duplicates found]
        LOG[Merge audit<br/>who merged what, when]
      end

      P((THE PERSON<br/>or ORG<br/>one record))

      P --- CP
      P --- EX
      P --- PR
      P --- AFF
      P -.→. PAR
      P --- PC
      P --- LOG
      RULE --> CAND
      P --- CAND

      subgraph business["Every product surface reads from the same record"]
        AU[Employees & Auth]
        WI[Leads, Deals, Cases,<br/>Agent Runs]
        ENR[Course & Batch<br/>enrolments]
        WAC[WhatsApp inbox]
        ACT[Timeline<br/>who did what]
        CAL[Calendar &<br/>invitations]
      end

      P -.→. AU
      P -.→. WI

```

```

P -. -> ENR
P -. -> WAC
P -. -> ACT
P -. -> CAL

```

```

classDef core fill:#312e81,stroke:#4338ca,color:#fff,stroke-width:3px
  classDef ident fill:#eef2ff,stroke:#6366f1,color:#1e1b4b
  classDef stateBox fill:#f0fdf4,stroke:#16a34a,color:#14532d
  classDef consentBox fill:#fef3c7,stroke:#d97706,color:#78350f
  classDef mergeBox fill:#fce7f3,stroke:#db2777,color:#831843
  classDef biz fill:#f8fafc,stroke:#94a3b8,color:#334155

```

```

      class P core
      class CP,EX ident
      class PR,AFF,PAR stateBox
      class PC consentBox
      class RULE,CAND,LOG mergeBox
      class AU,WI,ENR,WAC,ACT,CAL biz

```

Read this diagram as: every person or organization gets one row at the centre. Everything the business does — sales, service, learning, messaging, calendar — points at that row instead of holding its own copy.

2. Six business outcomes this foundation delivers

Outcome	What advisors and managers see	What it prevents
One screen per person	A lead's full history: past enrolments, WhatsApp conversations, cases raised, consent state, everything one click away	Multi-tab hunts across pipeline, learners, cases, inbox
Multi-channel reachout done right	Choose which of a person's numbers or emails to use; know which channel they've opted in on	Sending a WhatsApp to their landline; missing that they gave a work + personal email
B2B ready	Track one buyer with 5 stakeholders at Acme, each with their own role	Storing five separate leads for one company decision
Trainer becomes learner	A trainer who takes an advanced course is the same person on both sides	Two disconnected records; missed cross-sell
DPDP / GDPR-defensible marketing	Every WhatsApp broadcast recipient has proof of opt-in on file. Unknown consent = blocked.	Regulator complaints, TRAI/EU fines
Duplicate leads caught automatically	Same person from Instagram Monday + referral Friday → shows in a review queue for one-click merge	Two advisors calling the same person; wasted 30–40 advisor-hours per month at current volume

3. How the model works — a person's journey

Below is a real customer journey: someone lands on a form, becomes a lead, gets contacted, opts into WhatsApp, converts to a learner, and later an AI agent updates their record. Notice one thing throughout: **their record ID never changes**.

```

sequenceDiagram
    autonumber
    participant Web as Landing form
    participant API as CRM
    participant DB as Customer record
    participant Advisor
    participant Agents as AI agents

    Web->>API: Prospect fills out form
    API->>DB: Create ONE customer record
    API->>DB: Store email as a contact method
    API->>DB: Store phone as a contact method
    API->>DB: Mark role: "Lead"

    Note over DB: They are a lead

    Advisor->>API: Rates them "hot"
    Advisor->>API: Records: they said yes to WhatsApp
    API->>DB: Consent captured with source + timestamp

    Note over DB: Consent on record

    Advisor->>API: Converts them to a learner
    API->>DB: End-date the "Lead" role
    API->>DB: Add a "Learner" role – same record

    Note over DB: Same person, new role

    Agents->>API: Outreach Agent drafts email
    API->>DB: Timeline entry: "Agent drafted an email"
    API->>DB: Audit log: full trace of the AI action

    Note over DB: Every action attributed

```

Key property: the customer's identity in our system stays constant from the moment they show interest until years later. Nothing has to be "converted" or "migrated" — the same record just accumulates roles and history.

4. What we borrowed — inspiration from the best in the industry

We didn't invent this from scratch. We picked the structural patterns from **ServiceNow** (the platform that runs enterprise service management at Fortune 500 scale) and the workflow patterns from

Salesforce (the world's largest CRM).

```

    flowchart LR
      subgraph SN["ServiceNow's approach"]
        SN1[Every human is<br/>one record<br/>employees + customers]
        SN2[One shared spine<br/>for leads, deals, cases]
        SN3[Roles pivot over time<br/>never delete history]
        SN4[Contact methods<br/>as their own records]
        SN5[System user<br/>for automated actions]
        SN6[Frozen snapshots<br/>on regulatory records]
      end

      subgraph SF["Salesforce's approach"]
        SF1[Multi-org membership<br/>one person, many companies]
        SF2[Duplicate detection<br/>as first-class feature]
        SF3[Merge audit<br/>reversible if needed]
        SF4[Marketing consent<br/>per channel]
        SF5[Company hierarchy<br/>parent / subsidiary]
      end

      subgraph DE["What Digital Edify shipped"]
        DE1[Unified person record<br/>employee or customer]
        DE2[Single work spine<br/>lead → deal → case → onboarding]
        DE3[Temporal roles<br/>full journey preserved]
        DE4[Multi-channel contact<br/>+ per-channel consent]
        DE5[System actor for<br/>every AI agent action]
        DE6[Immutable case history<br/>at time of creation]
        DE7[Person↔org affiliations<br/>+ org hierarchy]
        DE8[4 matching strategies<br/>run continuously]
        DE9[One-click merge<br/>with full audit trail]
        DE10[DPDP/GDPR-strict<br/>WhatsApp + email consent]
      end

      SN1 --> DE1
      SN2 --> DE2
      SN3 --> DE3
      SN4 --> DE4
      SN5 --> DE5
      SN6 --> DE6

      SF1 --> DE7
      SF2 --> DE8
      SF3 --> DE9
      SF4 --> DE10
      SF5 --> DE7
  
```

```
classDef sn fill:#e0f2fe,stroke:#0369a1,color:#0c4a6e
```

```

classDef sf fill:#f0fdf4,stroke:#16a34a,color:#14532d
classDef de fill:#eef2ff,stroke:#4338ca,color:#312e81

class SN1,SN2,SN3,SN4,SN5,SN6 sn
class SF1,SF2,SF3,SF4,SF5 sf
class DE1,DE2,DE3,DE4,DE5,DE6,DE7,DE8,DE9,DE10 de

```

4.1 What ServiceNow does best — and we adopted it

ServiceNow is what runs support desks at Cisco, Bank of America, and Deloitte. Their core insight is: *stop separating people by role. Just have one table with one row per human.*

Their pattern	How we do it
One user table for all humans — employees, customers, external contacts	One customer record for every person. A flag tells us if they're an employee.
Every task (incident, change, request) shares one parent structure	Every work item (lead, deal, case, onboarding, agent run) shares one parent structure
A caller who becomes a customer keeps the same ID — a role is added, not swapped	A lead who converts to a learner keeps the same record — role toggles, history intact
Contact channels live as separate records — phone here, mobile there, email another	Contact methods are records, not fields. One person can have many.
A "System" account owns automated changes so audit logs stay clean	Every AI agent's action is attributed to a "System" record per tenant
Case records freeze the requester's identity at creation time (legal / regulatory need)	Same — our support cases keep an immutable snapshot

4.2 What Salesforce does best — and we adopted it

Salesforce is the largest CRM on the planet. Their strengths are workflow-heavy: how you find and clean duplicate records, how you track a buying committee, how you record consent.

Their pattern	How we do it
Multi-affiliation — one person can be at multiple companies with multiple roles	Our person↔org affiliations table, temporal (validity dates), one primary org per person
Duplicate Rules + Matching Rules — configurable, run automatically	4 default matching rules per tenant, plus a scanner that runs every 6 hours and on demand
DuplicateRecordSet + merge audit	Every merge writes a full snapshot of what got reparented, who did it, and when
Account.ParentId for company hierarchy	Same — one record can be marked as a subsidiary of another
Consent per marketing channel (via Marketing Cloud)	Consent per channel is a first-class table with opt-in state, source, evidence link, and history

4.3 Where we went further than either

Two places where the standard patterns weren't enough for our regulatory context:

- **Strict-block consent enforcement.** Salesforce treats consent as advisory — you can send to anyone unless they explicitly opted out. We do the opposite: **unknown consent is blocked**. This matches how DPDP and GDPR actually treat consent (require proof, not just absence of objection).
- **Merge is fully reversible via audit.** ServiceNow's merge doesn't preserve enough to unwind. We snapshot the losing record and every FK edge that got reparented, so an operator with the audit row can hand-reverse a bad merge if needed.

5. Duplicate → merge — how the review queue works

One of the most-quoted advisor pain points is *"we've called the same lead twice"*. Here's how the model catches it before it happens.

```

                                flowchart TB
    A[Same person arrives twice<br/>Instagram Monday · referral Friday] --> B[{Scanner ru
    B -->|Match rule fires<br/>same email OR<br/>same phone OR<br/>fuzzy name+city| C[Dup
                                C --> D{Manager reviews}
                                D -->|Not a duplicate| E[Dismiss<br/>candidate closed]
                                D -->|Yes, merge them<br/>one click| F[Preflight safety checks]
    F -->|Rejects merging two<br/>employees automatically| K[Blocked · human decides]
    F -->|Clean pair| G[All history moves to winner:<br/>contacts, activities, cases,<br/
    G --> H[Loser marked merged<br/>points at winner<br/>never deleted]
    H --> I[Full audit trail:<br/>who, when, what moved]

    classDef good fill:#f0fdf4,stroke:#16a34a
    classDef bad fill:#fef2f2,stroke:#dc2626
    classDef process fill:#eef2ff,stroke:#4338ca
                                class E,K bad
                                class G,H,I good
                                class B,C,D,F process
  
```

The four matching strategies (each configurable per tenant, all enabled by default):

1. **External system ID match** — the same person came in via two forms but Instagram or Razorpay tags them the same. Highest confidence.
2. **Exact email match** — same email on two records.
3. **Phone match (E.164)** — same normalized phone number across records.
4. **Fuzzy name + same city** — misspellings, "Priya N." vs "Priya Nair" in Bangalore. Requires review.

What happens on merge: the winner keeps their record ID. Every contact method, external ID, role, consent record, activity, work item, WhatsApp conversation, calendar invite, and audit trail from the loser gets reparented to the winner in one transaction. The loser is marked merged (not deleted) and points at the winner so any bookmark or link still works.

Safety rails: two employees cannot be merged automatically. Two records where one has an authenticated login and the other doesn't will merge cleanly. Everything is preflighted before any data moves.

6. Consent enforcement — how DPDP / GDPR compliance works

Every marketing message that leaves the platform is gated by explicit, dated proof of consent.

```

flowchart TB
    UI[Ops adds 500 people<br/>to a WhatsApp broadcast] --> Route[Broadcast recipient han]
    Route --> Filter[Check consent for<br/>WhatsApp channel<br/>on each person]
    Filter --> Query{Their current consent}
    Query -->|Opted in| Allowed[Added as recipient]
    Query -->|Opted out| Blocked1[Skipped – reason: opt-out]
    Query -->|No record at all| Blocked2[Skipped – reason: no consent<br/>strict block]
    Allowed --> Insert[Recipient inserted]
    Blocked1 --> Response
    Blocked2 --> Response
    Insert --> Response[Ops sees:<br/>N added, M skipped<br/>with reason for each]

    classDef strict fill:#fef2f2,stroke:#dc2626
    classDef ok fill:#f0fdf4,stroke:#16a34a
    class Blocked1,Blocked2 strict
    class Allowed,Insert ok
  
```

What the operator sees when adding recipients:

Added 342 · Skipped 158 — 47 opt-out, 111 no consent

They can then click on the skipped list, see who was blocked and why, and either update their consent (if the person just gave it verbally, for example) or export the list for a re-consent campaign.

Why "unknown = blocked" matters: DPDP and GDPR both require *proof of consent*, not just *absence of objection*. A lead who filled out a form 6 months ago and never explicitly opted into WhatsApp cannot legally be broadcast to. Our system blocks by default — the operator has to affirmatively grant consent before that person can be reached on that channel.

Every consent change writes an activity row to the person's timeline — so a regulator asking "when did this person opt in, and what did they see?" gets an audit trail with source (signup / unsubscribe / legal request), date, and evidence URL.

7. Side-by-side comparison

If you know ServiceNow or Salesforce, this table shows exactly which patterns we inherit and where we differ.

Concern	ServiceNow	Salesforce	Digital Edify
One person table	Yes (sys_user)	No (two tables, bridged)	Yes — ServiceNow-style
Universal work spine	Yes (task parent)	No (separate objects)	Yes — ServiceNow-style
Lead → Customer transition	Role added, no data lost	Destructive (Lead deleted)	Role added — ServiceNow-style
Company hierarchy	Yes	Yes	Yes
Multi-affiliation	Limited	Yes	Yes — Salesforce-style
Contact methods as records	Yes	Partial	Yes — hybrid
Deduplication engine	Yes (CIRE)	Yes (Duplicate Rules)	Yes — Salesforce-style
Merge audit trail	Basic	Yes	Yes with full snapshot — beyond both
Consent per channel	Weak in core	Marketing Cloud (extra product)	First-class + DPDP/GDPR strict
System actor for automation	Real user	Hidden user	Dedicated sentinel record
Multi-tenancy at DB level	Namespace	Namespace	Row-level security in Postgres

8. What's live right now

After a fresh install, the platform starts with:

- 1 tenant (Digital Edify)
- 23 records total: 1 system sentinel, 10 internal users (admin, advisors, service reps, super-user), 12 seeded leads
- 34 contact methods (email + phone per person, WhatsApp for inbox contacts)
- 4 matching rules ready to run
- 7 seeded consent rows (3 opt-in, 1 opt-out, rest unknown to demo the strict-block behaviour)
- 60 timeline entries — every one traceable to a specific actor (advisor, admin, or the System agent)

And it stays that way: every new lead, every AI agent action, every WhatsApp inbound message goes through the same path — one record for the person, one contact method per channel, one consent state, and every action attributed via audit trail.

9. Business impact — measurable

Metric	Before	After
Duplicate leads caught	0 (advisor memory)	~5–10 caught per week by rule scanner + on-demand
Time to answer "show me everything about this person"	4–5 tabs, 60–90 seconds	One screen, instant
WhatsApp broadcast regulatory exposure	Every send at risk	Every send provably consented
B2B readiness	Not modelled	Full stakeholder graph
Reversible merge audit	None	Full snapshot in every merge log
Advisor-hours saved per month (est., current volume)	—	30–40

10. Where the industry is going, and where this positions us

The trend across enterprise SaaS is toward **one identity per person, with every product surface pulling from it**. Microsoft Dynamics moved this way in 2022. HubSpot rebuilt their contact model on similar principles in 2023. Salesforce's Data Cloud (2023) explicitly names their target the "single customer view."

We shipped the foundation before we needed it, which means:

- **Every future product surface** (finance, career services, alumni network, referrer program) plugs in with zero data-model rework
- **AI agents** can reason across the full customer graph without stitching identities together at query time
- **Compliance work** (DPDP audits, ISO reviews, SOC 2) has a clean data-lineage story to tell
- **Enterprise sales** (selling into orgs that expect ServiceNow / Salesforce-grade data models) can be answered on merit

Appendix — Glossary in plain English

Term	Plain meaning
Party	The single record for any person or organization — customer, employee, alumnus, referrer, vendor
Role	A capacity in which a person participates — lead, learner, advisor, alumnus. A person can hold many roles over time.
Contact method	A specific way to reach a person — one email, one phone number, one address. A person has many.
Affiliation	A person's membership in an organization, with their role there and validity dates
Consent	A recorded permission to contact this person on a specific channel, with date and source
Match rule	A configurable strategy for detecting that two records are the same person
Merge	Combining two records into one, preserving all history
Sentinel	A special "System" record used to attribute AI agent and system-automated actions in the audit trail
DPDP	India's Digital Personal Data Protection Act, 2023
GDPR	The EU's General Data Protection Regulation
Row-level security	Postgres feature that enforces tenant isolation at the database level — one tenant literally cannot query another tenant's rows